University College London, 2025-2026
Econ 0016 – International Macroeconomics – Franck Portier

Problem set 2 : Terms of Trade, the World Interest Rate, Tariffs, and the Current Account – Current Account Determination in a Production Economy

Chapter 4 and 5 in Schmitt-Grohé, Uribe and Woodford

Version 1.0 - 10/01/2026

True/False/Uncertain

1. In an endowment small open economy, the income effect of an increase in the world interest rate is positive if the country NIIP is negative.
2. A country populated by more impatient households (i.e., a country with a lower β) will consume more and invest less in period 1.
3. Good news about future productivity increases consumption already now regardless of whether the economy is open or closed.

Exercise 4.7 (Import Tariffs)

Consider a two-period open economy in which households have preferences given by

$$\log C_1 + \log C_2,$$

where C_1 and C_2 denote consumption of food in periods 1 and 2, measured in tons. The country does not produce food. Households are endowed with Q_1 and Q_2 barrels of oil in periods 1 and 2. In both periods, a barrel of oil sells for one ton of food in international markets. The economy starts period 1 with no assets, $B_0 = 0$. The world interest rate is $r^\star$ and there is free capital mobility. The government imposes tariffs on food imports in periods 1 and 2, denoted τ_1 and τ_2, and rebates the revenue generated by the tariffs to the public using lump-sum transfers, denoted L_1 and L_2.

1. What are the terms of trade in periods 1 and 2?
2. Derive the household's intertemporal budget constraint.
3. Write down the optimization problem of the household.
4. Derive the first-order conditions of the household's optimization problem.
5. Write down the budget constraints of the government in periods 1 and 2.
6. Combine the household's intertemporal budget constraint with the government budget constraints to find the economy's intertemporal resource constraint in equilibrium. Do the policy variables τ_1, τ_2, L_1, or L_2 appear in this constraint? Why?

7. Let $Y \equiv Q_1 + Q_2/(1+r^\star)$ denote the present discounted value of the endowment path. Express the equilibrium values of consumption in periods 1 and 2 in terms of Y, $r^\star$, τ_1, and τ_2.

8. Write the equilibrium trade balance in period 1 in terms of Y, $r^\star$, Q_1, τ_1, and τ_2. Compare the trade balance under free trade, i.e., $\tau_1 = \tau_2 = 0$, to the following cases: (a) $\tau_1 = \tau_2 > 0$; (b) $\tau_1 > 0$ and $\tau_2 = 0$; and (c) $\tau_1 = 0$ and $\tau_2 > 0$.

9. Define $x = \frac{1+\tau_1}{1+\tau_2}$. Use the equilibrium values of C_1 and C_2 derived in question 7 to eliminate these two variables from the household's utility function. Find the value of x that maximizes the household's welfare. Interpret your result.

Exercise 5.5 (An Open Economy with Investment)

Consider a two-period model of a small open economy with a single good each period. Let preferences of the representative household be described by the utility function

$$\log C_1 + \log C_2,$$

where C_1 and C_2 denote consumption in periods 1 and 2. Each period the household receives profits from the firms it owns, denoted Π_1 and Π_2. Households and firrmshave access to financial markets where they can borrow or lend at the interest rate r_1. The production technologies in periods 1 and 2 are given by

$$Q_1 = A_1 I_0^\alpha$$

and

$$Q_2 = A_2 I_1^\alpha,$$

where Q_1 and Q_2 denote output in periods 1 and 2, I_0 and I_1 denote the capital stock in periods 1 and 2, A_1 and A_2 denote the productivity factors in periods 1 and 2, and α is a parameter between o and 1. Assume that $I_0 = 16$, $A_1 = 3 + \frac{1}{3}$, $A_2 = 3.2$, and $\alpha = 0.75$. At the beginning of period 1, households have $B_0^h = 8$ bonds. The interest rate on bonds held from period 0 to period 1 is $r_0 = 0.25$. In period 1, firms borrow the amount D_1^f to purchase investment goods that become productive capital in period 2, I_1. Assume that there exists free international capital mobility and that the world interest rate, denoted $r^\star$, is 20 percent.

1. Compute output and profits in periods 1 and 2.

2. Compute the optimal levels of investment in period 1 and output and profits in period 2.

3. Solve for the optimal levels of consumption in periods 1 and 2.

4. Find the country's net foreign asset position at the end of period 1, denoted B_1,saving, S_1, the trade balance, TB_1, and the current account, CA_1.

5. Now consider an interest rate hike in period 1. Specifically, assume that as a result of turmoil in international financial markets, the world interest rate increases from 20 percent to 50 percent in period 1. Find the equilibrium levels of saving, investment, the trade balance, the current account, and the country's net foreign asset position in period 1. Provide intuition.

6. Suppose that the interest rate is 20 percent, and that A_1 increases to 4. Calculate the equilibrium values of output, consumption, saving, investment, and the current account in period 1. Provide an intuitive interpretation of the adjustment to the transitory productivity shock.

7. Suppose that the interest rate is 20 percent, that $A_1 = 3 + \frac{1}{3}$, and that A_2 increases from 3.2 to 4. Calculate the equilibrium values of consumption, saving, investment, and the current account in period 1. Explain your findings.